

Q1. Laplace Transforms

The complex frequency domain expression for a current signal is

$$I(s) = \frac{5}{s^2} - e^{-5s} \frac{5}{s^2} + \frac{5}{s^2 + 2.5 \times 10^5} \quad (1)$$

What is the time domain current signal expression?

Q2. Laplace circuit analysis

- (a) The series RC circuit in Figure 1 has a source $v(t) = 5.0u(t)$. Using the Laplace transform, determine the voltage signal in the time domain across the capacitor, and the current signal in the time domain in the circuit.

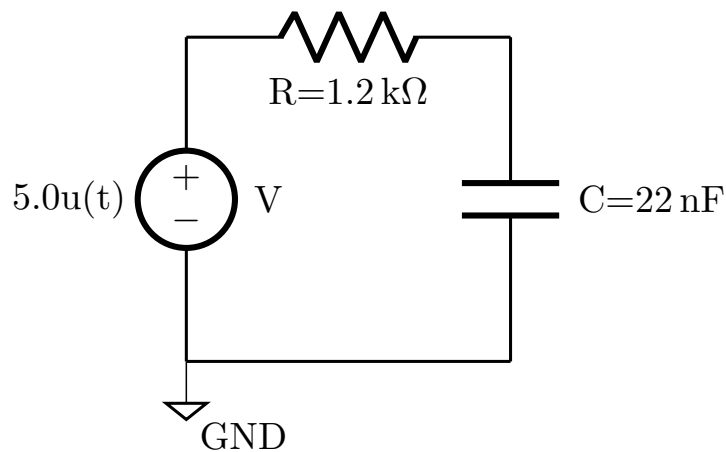


Figure 1:

- (b) The parallel RL circuit shown in Figure 2 has a voltage source $v(t) = 6tu(t)$. Use The Laplace transform to determine the current signal in the time domain through the inductor.

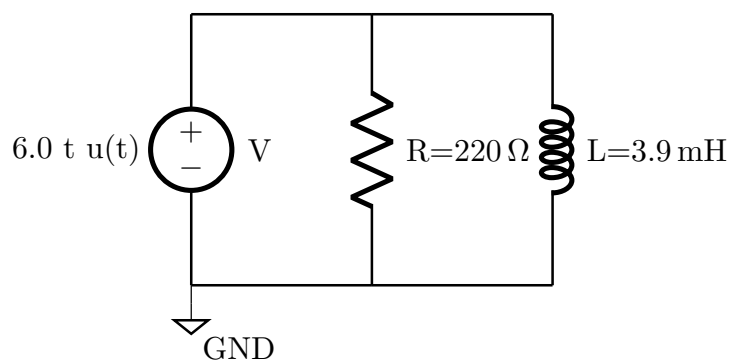
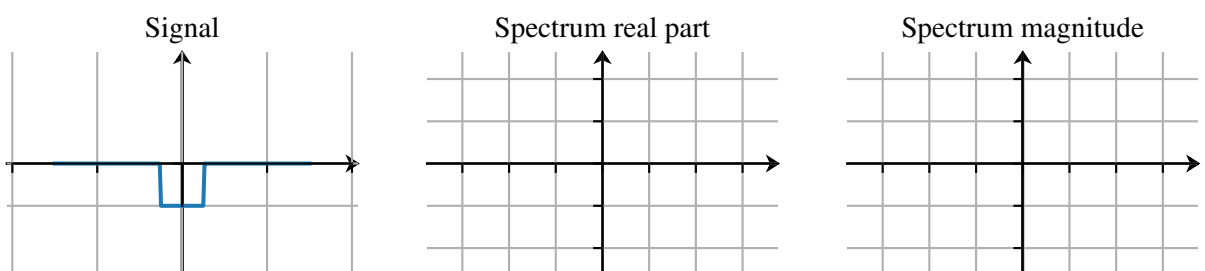
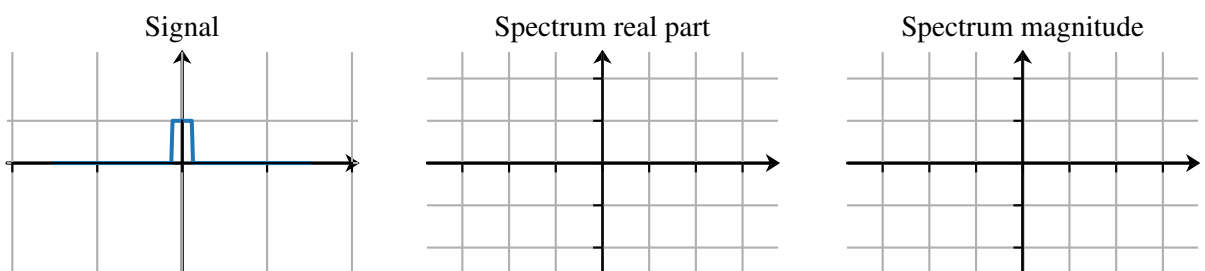
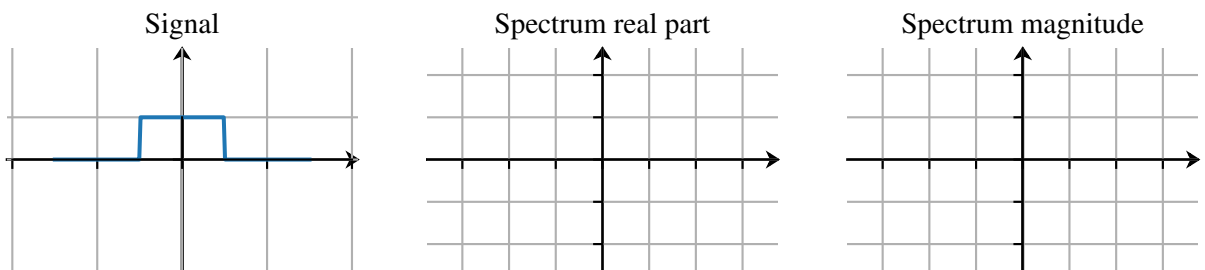
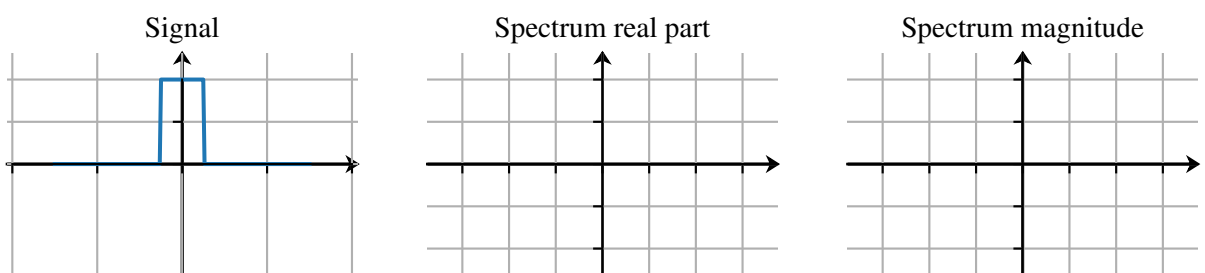
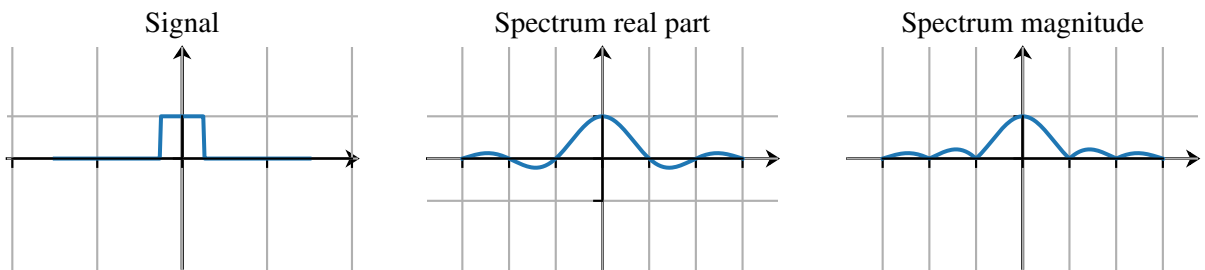


Figure 2:

Q3. Fourier Transforms

Complete the following Fourier transform pairs:



Q4. Fourier Transforms 2

For the following time-domain signals, derive the Fourier transform using the common Fourier transform pairs in theorems in the Formula Sheet. Sketch the signals, and the magnitude of their spectra.

- (a) $\text{rect}(2t)$
- (b) $\text{rect}(t/2)$
- (c) $\text{sinc}(2t)$
- (d) $\text{sinc}(t/2)$
- (e) $1 + \cos(2\pi f_0 t)$
- (f) $\text{rect}(t) + \text{sinc}(t)$
- (g) $\exp(j2\pi f_0 t)$
- (h) $\delta(t - \tau)$
- (i) $\text{rect}(t - \tau)$
- (j) $\text{rect}(t/T) \cos(2\pi f_0 t)$

Q5. Inverse Fourier Transforms

For the following spectra, derive the inverse Fourier transform using the common Fourier transform pairs and basic theorems in the Formula Sheet.

- (a) $\text{rect}(2f)$
- (b) $\text{rect}(f/2)$
- (c) $\text{sinc}(2f)$
- (d) $\text{sinc}(f/2)$
- (e) $\delta(f - 2) + \delta(f + 2)$
- (f) $\delta(f - 2) - \delta(f + 2)$
- (g) $\delta(f - 2)$